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SECP3843 - SPECIAL TOPIC IN DATA ENGINEERING

SECTION 2

MS AZURE TUTORIAL REPORT

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1.0 Project Overview

Problem Background

In the modern data landscape, organizations struggle to unify fragmented data sources into actionable insights. Traditional on-premise data systems often lack the scalability and agility required to process large-scale datasets efficiently. This project addresses the challenge of building a robust, cloud-native infrastructure that can automate the journey of data, from its raw, unstructured state to a polished, visual format without the overhead of maintaining physical hardware.

Project Objectives

The primary goal of this project is to implement a fully automated data pipeline using Microsoft Azure. The specific objectives include:

- Data Ingestion: Establishing a secure connection to ingest raw data from external sources into a centralized cloud storage environment.
- Pipeline Automation: Leveraging Azure Data Factory to orchestrate seamless data movement and transformation.
- Data Transformation: Processing and cleaning data to ensure it is structured and ready for analytical consumption.
- Actionable Analytics: Developing an MS Power BI dashboard to provide meaningful visualizations and data storytelling for business stakeholders.

Business and Industry Relevance

This solution mirrors real-world industry requirements where data engineering is the backbone of business intelligence. By utilizing a "Bronze, Silver, and Gold" data architecture in the tutorial, businesses can ensure data integrity and high-speed reporting. Implementing this on Microsoft Azure demonstrates a cost-effective, scalable approach to data management that allows companies to make data-driven decisions in real-time, providing a significant competitive advantage.

Project Scope

The scope of this project encompasses the end-to-end development of a data pipeline. It begins with raw data storage in Azure Data Lake, coding for silver and gold data processing in Azure Databricks, moves through building the data pipeline architecture via Azure Data Factory, and concludes with the deployment of an interactive Power BI reporting layer. The project is limited to the functionalities provided within the MS Azure environment.

2.0 Description on The Solution

An end-to-end cloud-based data engineering pipeline built with Microsoft Azure services is the solution created in this tutorial demonstrating. In order to ensure effective data processing and analysis, the system is built to automate every stage of the data lifecycle from data ingestion to final visualization.

The implementation follows the Medallion Architecture, consisting of Bronze, Silver and Gold layers. Raw data files were absorbed at the Bronze layer and kept in their original format in Azure Data Lake Storage. This layer acts as the first landing zone, guaranteeing the unaltered preservation of all source data.

Next, the Silver layer is responsible for data cleaning, transformation and integration. Azure Databricks is used to process raw data in order to apply business logic, handle missing information and eliminate inconsistencies. This step guarantees that the data is consistent, organized and prepared for additional analysis.

The final, analytics-ready dataset is represented by the Gold Layer. Azure SQL Database stores the processed data in an organized format, usually with reporting and query-optimized schemas. This makes it possible to obtain high-quality data quickly and effectively. Azure Data Factory orchestrates the pipeline in order to automate the workflow. It controls the flow of data between layers and guarantees that every operation operates on time and dependably.

Lastly, the visualization layer is connected with Microsoft BI. It allows users to make data-driven decisions by connecting to the Gold layer data and displaying insights through interactive dashboards. All things considered, this solution shows how to use cloud technologies for current data engineering in a scalable, automated and effective manner.

3.0 Technology Used

This project leverages the Microsoft Azure ecosystem to build a scalable, cloud-native data engineering solution. Each service was selected for its ability to handle specific stages of the data lifecycle, ensuring high availability and seamless integration.

- Azure Data Lake Storage (ADLS) Gen2 Storage Account: Serves as the primary storage layer for the project. It is used to implement a tiered storage architecture where we store Bronze, Silver, and Gold folders as containers, allowing for the separation of raw, cleansed, and curated datasets.
- Azure Data Factory (ADF): Acts as the central orchestration engine. It is used to create and manage data pipelines that automate the movement of data from the source to the storage layers through "Copy Data" activities and "Triggers". The medallion architecture is applied using this application.
- Azure SQL Database: Provides a relational storage environment for the final, curated data. This service hosts the "Gold" layer data, making it easily accessible for high-performance querying and reporting.
- Azure Key Vault: Used to enhance security by securely storing and managing secrets, such as database connection strings and access keys, ensuring that sensitive information is not hard-coded within the ADF pipelines. It is used to store all the sensitive keys and passwords that are used in this tutorial like the SQL Server Management Studio password, username and the databricks key.
- Azure Databricks Service: Employed for advanced data transformation and cleaning. Using Python language notebooks in Databrick, data is processed at scale to handle transformations from bronze to silver and silver to gold.
- Azure Synapse: Used as the primary orchestration engine to automate the end-to-end workflow. We implemented dynamic pipelines using Get Metadata and ForEach activities to programmatically iterate through data layers, ensuring that the creation of SQL views remains automated and scalable as new datasets are introduced. It uses a Serverless SQL pool to create views over the Gold Delta files, allowing Power BI to query the data using standard SQL without the need for a dedicated database.
- Microsoft Power BI: Used for the final visualization layer. It connects directly to the Azure SQL Database to transform curated data into interactive dashboards for business storytelling.
- Azure Active Directory (Entra ID): Utilized to provide a secure identity framework. By using Managed Identities, eliminate the need for hardcoded credentials in the pipeline, and RBAC roles were assigned to ensure that only authorized services could access the sensitive data stored in the Bronze, Silver, and Gold layers.

4.0 Data Pipeline Architecture

The architecture follows a Medallion Architecture implemented on the Microsoft Azure Cloud. This design ensures data integrity as it moves from raw on-premises sources to a refined state ready for business intelligence. The architecture diagram applied in this tutorial is shown in Figure 1.

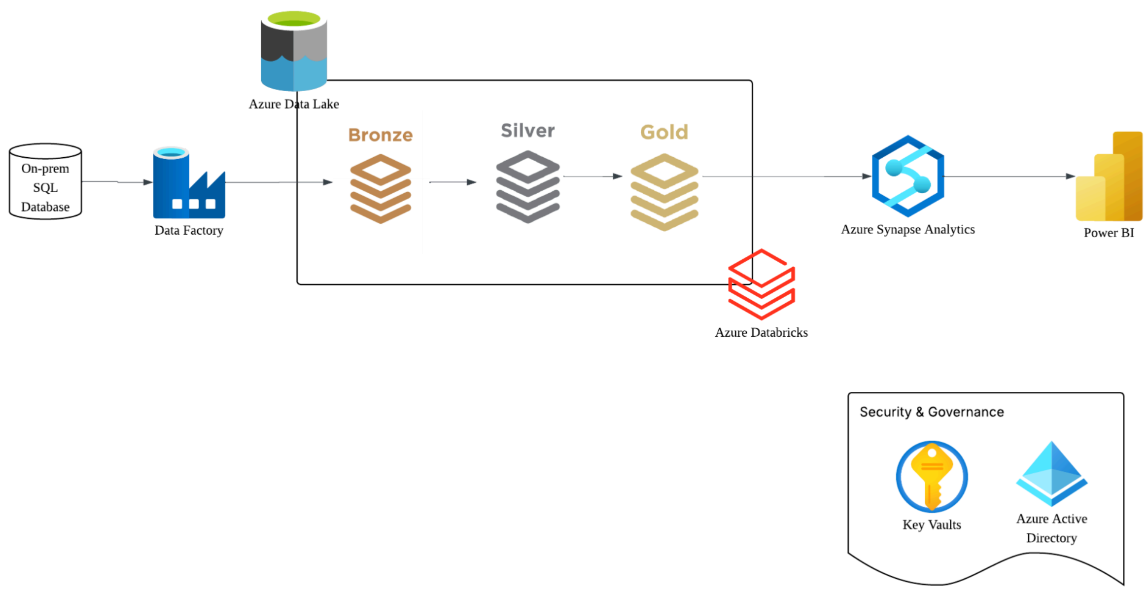

Figure 1: Azure Tutorial Data Pipeline Architecture

1. Data Ingestion (On-Prem to Bronze)

- Source: The pipeline begins with an On-premises SQL Server (SSMS).
- Ingestion Tool: Azure Data Factory (ADF) acts as the orchestration engine. It utilizes a Self-Hosted Integration Runtime (SHIR) to securely bridge the gap between the local private network and the Azure cloud.
- Destination (Bronze Layer): Data is landed in its rawest form in Azure Data Lake Storage Gen2 (ADLS Gen2). This "Bronze" layer acts as a historical record, preserving the original state of the source data in Parquet format.

2. Data Transformation (Silver and Gold)

- Processing Engine: Azure Databricks is utilized for its high-performance Spark clusters.
- Silver Layer (Cleansing): Databricks notebooks read from the Bronze layer to perform data cleansing. This includes handling null values, correcting data types, and implementing the snake_case naming convention for standardizing column headers.

- Gold Layer (Curated): The cleaned data is then transformed into the Gold layer. Here, the data is aggregated and structured into business-level entities (Fact and Dimension tables), optimized for reporting performance and stored in Delta format for ACID compliance.

3. Data Serving and Visualization

- Serving Layer: Azure Synapse Analytics provides a Serverless SQL Pool that acts as a relational "view" over the Gold Delta files. This eliminates the need to move data into a heavy database, reducing costs while providing a standard SQL endpoint.
- Visualization: Power BI connects to the Synapse endpoint. This allows for the creation of interactive dashboards where stakeholders can derive insights from the processed data.

4. Security and Governance

- Azure Key Vault: To adhere to security best practices, sensitive credentials like SQL passwords and storage keys are never hardcoded. They are stored as secrets in Key Vault and retrieved dynamically by ADF.
- Azure Active Directory: Provides the security backbone through Identity and Access Management (IAM) and Managed Identities, ensuring that only authorized services can communicate across the pipeline.

5.0 MS Power BI Dashboard

Using data from an Azure-based data pipeline, the Power BI dashboard created for this project provides a comprehensive summary of sales performance. With slicers and filters, the dashboard's essential visual components such as total sales, product category breakdown, and product-level analysis allow for interactive data exploration.

Total Sales, a crucial KPI in the dashboard, is computed dynamically using a DAX measure based on order quantity and unit pricing from the SalesOrderDetail table. This guarantees that, following a data refresh, any database adjustments are instantly reflected in the visualization layer.

The total sales value was roughly 700.69K prior to the insertion of a new sales record. The Total Sales rose to about 714K when a new transaction was added to the SQL database and the Power BI dataset was refreshed. This shows that the dashboard can reflect changes made in real time via the data pipeline and is correctly connected to the underlying data source.

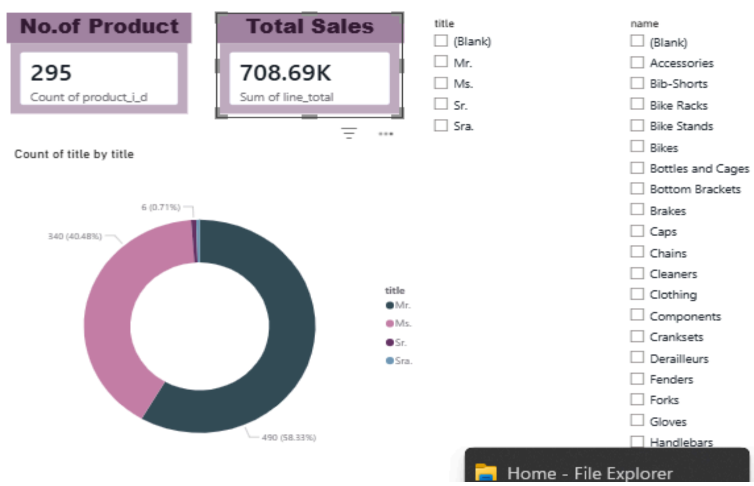

Figure 2: Dashboard with Original Data

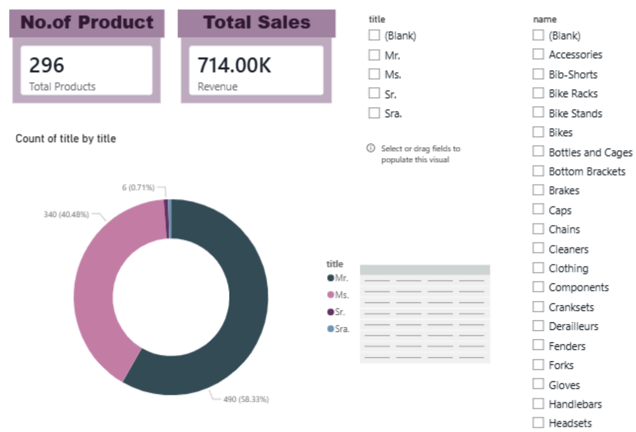

Figure 3: Dashboard with Updated Data

Additionally, the dashboard has interactive slicers for product title and category that let users dynamically filter sales data. Additionally, visual elements like donut charts and cards offer both comprehensive and concise views on product distribution and sales contribution.

Activities Performed in Power BI Dashboard Development:

Several crucial tasks were completed throughout the dashboard's development:

1. Imported data from SQL Server (the AdventureWorksLT2025 database) that had been cleansed and organized.
2. Established connections between several tables, including Customer, SalesOrderDetail, Product, and ProductCategory.
3. Developed a paradigm that resembles a star schema to guarantee effective reporting and filtering.
4. To compute important metrics like Total Sales, DAX measures were developed.
5. Created interactive graphics, such as:
    - KPI cards (Product Count/Total Item Sold, Total Sales)
    - Donut charts (distribution of products)
    - Slicers (product and category screening)
6. By adding a new entry to the SalesOrderDetail table and verifying the modifications in Power BI following a refresh, data updates were tested.

6.0 Reflection

Damiya Aina binti Basir Abd Shammad

This project gave me hands-on experience utilizing Azure and Power BI to create a comprehensive data engineering and analytics pipeline.

First, I gained knowledge about how Azure Data Factory pipelines operate in practical situations, particularly with regard to the transfer and transformation of data between various tiers. I also realized how crucial proper configuration is, since little problems with permissions or data mapping can have an impact on the pipeline as a whole.

Second, I gained a deeper comprehension of Power BI, especially with regard to data modeling, relationships, and DAX computations. At first, I had issues where wrong fields and relationships caused visualizations to update incorrectly. Nonetheless, this enhanced my attention to detail and troubleshooting abilities.

Last but not least, this project helped me better grasp how business intelligence tools work with cloud-based platforms. I can see how important insights that aid in decision-making can be derived from unprocessed data. Overall, this project strengthened both my technical skills and problem-solving ability in data engineering and analytics.

Nurul Ika Syafiny Binti Azhar

Prior to this tutorial, I viewed cloud services as individual tools. Through this project, I have learned how to orchestrate a complex ecosystem. I gained hands-on experience in configuring a Self-Hosted Integration Runtime, which taught me the realities of connecting local, on-premises data to the cloud. I now understand how Azure Data Factory acts as the "heart" of the system, timing the movement of data so that Databricks and Synapse always have the most current information.

The most valuable part of this project was overcoming technical hurdles. For example, I encountered errors where "Test Connection" succeeded, but the actual pipeline run failed. This taught me that credential validation is only one step; permissions (IAM) and firewall rules are equally critical for data movement. Encountering "Unique Key Violations" in SSMS forced me to dive deeper into database constraints and the importance of using variables like @x for handling complex data types like varbinary.

Implementing the Bronze, Silver, and Gold layers changed my perspective on data quality. I learned that data is rarely "ready" when it arrives. By using Databricks to transform raw Parquet into cleaned Delta files, I saw firsthand how standardized naming conventions, snake_case and schema enforcement make the final data much easier to use in Power BI.

To conclude this reflection, this tutorial has taught me so much about the service offered in Azure and how I can apply it in future projects like the PPG case study project. Despite the errors I encountered along the way, it is worth the trouble for the knowledge I learned as a future data engineer and the experience I got.